Source-indexed information repair: stable representations, lawful acquisition, and online realization

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Abstract

We distinguish three finite mathematical problems: autonomously retaining future-task information, lawfully acquiring an original source label through possibly destructive probes, and realizing abstract requests online. Stable partition refinement gives a canonical deterministic repair and a sharp fixed-width auxiliary-memory bound. For acquisition, a source-indexed belief construction yields exact rational minimax values with locally checkable cost and termination certificates. The minimal retained advice at a given probe budget is characterized by weak coloring of the hypergraph of minimal infeasible source sets. A three-source example has all pairs identifiable at unit cost but no joint identification strategy. The general ingredients are classical; the contribution is a precise integration with explicit resource, provenance, and information boundaries, not a claim of established novelty.

1 Scope and mathematical interfaces

All carriers in the constructive results are complete finite declared models. An original source label records provenance as a mathematical parameter; it does not authenticate physical reality. All costs are nonnegative rationals. There are no probabilistic, asymptotic, physical-energy, or Riemann-hypothesis conclusions.

Our baseline is PR179, commit `81684b4b8956f9f5da56c376ba7e4339277eb665`. Sibling PR178, commit `e42096ec0186228dd0f748767599e470f2ca2e1d`, was inspected separately. The sibling's invariant private source tag differs from the baseline's preserved source output; their failed-run journals also differ. We reuse their common local kernel-congruence requirement, not an identification of their full semantics. In particular, acquisition below has no refusal oracle: all probes must be lawful throughout the current uncertainty set.

2 The least autonomous repair

Fix a nonempty finite state space X, output $o : X \to O$, and partial deterministic actions $\delta_a : D_a \to X$ with costs $c_a : D_a \to \mathbb{Q}_{\geq 0}$. Any immutable source identity or designated proof-task result that must be retained is included in o. Let $q : X \to Y$ be an existing representation. An equivalence is *stable* when it preserves output, enabledness, cost, and successor equivalence for every action. Let R_* be the greatest stable equivalence contained in $\ker q$.

Theorem 2.1 (Classical stable refinement and factorization)**.** *R_* exists and is obtained by finite partition refinement starting from* $\ker(q, o)$*. There are at most* $|X| - |(q, o)(X)|$ *strict refinements.*

If $r : X \twoheadrightarrow Z$ is any autonomous sufficient repair retaining q, there is a unique map $f : Z \to X/R_*$ such that $\pi_* = f \circ r$.

Proof. Refine by each action's unavailability or pair of cost and previous successor class. Agreement through k actions is exactly the kth partition, by induction. A plateau is stable. Every strict refinement increases the number of blocks, giving termination. Any stable equivalence refining (q, o) refines every stage by induction, hence refines the terminal partition. The kernel of r is such an equivalence. Therefore $f(r(x)) = [x]_{R_*}$ is well-defined and unique by surjectivity. □

This is enriched Moore/Nerode minimization, within classical coalgebraic semantics [1, 2]. It is not merely the intersection of $\ker q$ with the old behavioral equivalence: future values of q must be preserved as well. For example, let all old outputs be constant and let the only action send $u \mapsto u$, $v \mapsto w$, $w \mapsto w$, at zero cost. If $q(u) = q(v) \neq q(w)$, the old behavior is constant, but the repair separates all three states.

Theorem 2.2 (Sharp final auxiliary alphabet)**.** *Put $K = \max_y |q^{-1}(y)/R_*|$. The smallest alphabet for an auxiliary label $\alpha : X \to \{0, \dots, k-1\}$ making (q, α) an autonomous sufficient repair has size K. Its smallest fixed binary width is $\lceil \log_2 K \rceil$.*

Proof. Any repair kernel lies inside R_*. Within one q-fiber, distinct R_* classes therefore need distinct auxiliary labels. Conversely, enumerate the classes separately in each fiber and reuse the same alphabet between fibers. Then $\ker(q, \alpha) = R_*$. Binary capacity gives the ceiling formula. □

The encoder may inspect the source state, while the decoder knows q. This is an existence and retention bound, not a construction of an observation instrument. The supplemental label need not update autonomously without q. No smaller code is obtained by supposing an unperformed acquisition occurred.

3 Acquiring an original label through destructive probes

Let X_0 be the original source set, Z a finite carrier, $i : X_0 \to Z$ an initialization, and $t : X_0 \to T$ the target original repair class. A probe maps its domain in Z to receipt, exact cost, and new carrier. A belief is a nonempty set $B \subseteq X_0 \times Z$ with at most one carrier for each original source. A probe is lawful only if defined at every carrier in B. For a receipt/cost outcome $h = (\rho, c)$, its successor is

$$B_{a,h} = \{(x, d_a(z)) : (x, z) \in B,\ (\rho_a(z), c_a(z)) = h\}.$$

The original coordinate remains fixed even when current carriers merge. Success means all originals in the terminal belief have the same t-value. Policies must terminate on every possible original. Their cost is the maximum total charge over those originals.

Theorem 3.1 (Exact terminating acquisition cost)**.** *Let the reachable belief arena, stopped at success, have N states. Initialize $V_0 = 0$ at success and $V_0 = \infty$ elsewhere. At nonsuccessful beliefs set*

$$V_{n+1}(B) = \min\left(V_n(B), \min_{a \text{ lawful}} \max_{h : B_{a,h} \neq \varnothing} \{c(h) + V_n(B_{a,h})\}\right).$$

Then V_N is the exact optimum among terminating policies. A finite optimum is attained and rational.

Proof. Induction identifies V_n with optimal identification trees of height at most n. If a finite successful tree repeats a belief on a branch, replace its earlier occurrence by the later subtree. The source/current belief is equal, so the substituted policy is lawful and correct. Deleted costs are nonnegative; no original source pays more. Repeated deletion leaves height at most N. There are finitely many trees of this bounded height, proving attainment and the stated recurrence bound. □

Proposition 3.2 (Local optimality certificate)**.** *An exact value certificate assigns $L(B) \in \mathbb{Q}_{\geq 0} \cup \{\infty\}$. Success states have value zero and stop. Every lawful action obeys*

$$L(B) \leq \max_h \{c(h) + L(B_{a,h})\}.$$

At each finite nonsuccessful state a selected action obeys the reverse inequality, has finite-valued children, and strictly decreases a supplied nonnegative integer rank along every outcome. These conditions imply $L = V$.

Proof. Induction on any finite policy tree proves that L lower-bounds its worst cost. This also excludes a finite successful policy at an infinity-labelled state. Induction on the selected policy's decreasing rank proves termination and an upper cost bound L. The two inequalities give equality.

Such ranks exist for the optimal values. Call an action optimal when its cost-plus-optimal-child maximum is at most the parent's finite value. Form the goal attractor using only these actions. If finite-valued states remain outside, choose their minimum value m, then a state of value m with a shortest optimal tree. Its root action is optimal, so some child is still outside the attractor. Nonnegative cost and minimality force that child's value to be m and its entering cost to be zero. Its subtree has cost at most m, hence is optimal and shorter, a contradiction. Attractor ranks therefore supply one proper positional optimal policy. This argument does not assume that every branch of a root-optimal tree is locally optimal. □

A zero-cost nonidentifying loop satisfies the spurious Bellman equation $v = 0 + v$ with $v = 0$; it fails the rank condition. Thus an arbitrary fixed point is not a termination certificate. The synthesis makes no claim that destructive acquisition restores the carrier for later task execution. That conclusion needs a separate source-preserving restoration or simulation map. Full transcript/controller memory is available during acquisition.

4 The hypergraph of acquisition obstructions

For budget C and original observation y, let $\mathcal{F}_C(y)$ be the nonempty subsets of $q^{-1}(y)$ identifiable within cost C. Restricting a policy to fewer originals preserves lawfulness and cannot increase worst-case cost. Consequently $\mathcal{F}_C(y)$ is hereditary and contains all singletons. Let $\mathcal{H}_C(y)$ be its minimal infeasible subsets.

Lemma 4.1 (Finite hereditary families)**.** *A finite set is feasible in a hereditary family exactly when it contains no minimal infeasible subset. The minimum number of feasible blocks partitioning the vertex set equals the weak chromatic number of its minimal-infeasible hypergraph, where no hyperedge may be monochromatic.*

Proof. Every infeasible finite set contains a minimal one; a feasible set cannot contain an infeasible subset. A coloring is therefore valid exactly when each color class is feasible. □

Theorem 4.2 (Exact prior-advice/acquisition tradeoff)**.** *Suppose $q(x)$ and an auxiliary source label $\alpha(x)$ are already retained before probing. The minimum common advice alphabet permitting cost-C identification is*

$$K_{\mathrm{prior}}(C) = \max_{y} \chi(\mathcal{H}_C(y)), \qquad b_{\mathrm{prior}}(C) = \lceil \log_2 K_{\mathrm{prior}}(C) \rceil.$$

Proof. After observing (y, α), the whole joint fiber must be identifiable. Thus every advice color class must be feasible. Conversely, use a feasible policy for each color class of a valid coloring. Apply Lemma 4.1 within each q-fiber and reuse colors between fibers. The binary formula follows from alphabet capacity. □

Advice may split inside a final repair class. It must have been acquired or retained earlier; this theorem does not provide a free source oracle. Prior advice width, final autonomous repair width, and temporary controller memory are three distinct quantities. Budget breakpoints are among the finitely many exact initial-subset values. Exhaustive finite synthesis gives a precise frontier, without a polynomial-time claim.

A three-source obstruction and sharp frontier

Originals x, y, z share q and have distinct target labels. A unit-cost probe a distinguishes x but sends y, z to the same carrier u with the same receipt. A unit-cost probe b distinguishes z but merges x, y into carrier v. At u, v every probe loops with the same receipt. Each pair is identifiable at cost 1, but every first action from the triple has an irreversibly indistinguishable branch. No joint identification policy exists.

At budget 1 the unique minimal forbidden set is $\{x, y, z\}$. Its pairwise conflict graph has no edges, but its weak chromatic number is 2. Add a nondestructive identifying probe of cost 2 at x, y, z. The exact frontier is

$$(b_{\mathrm{prior}}, C) = (2, 0), \quad (1, 1), \quad (0, 2).$$

This proves that pairwise minimum distinguishing costs do not determine joint acquisition feasibility. Forgetting the original coordinate would falsely declare the merged carrier resolved and change the task itself.

5 Online source realization and public command interfaces

Offline history existence and online control have different quantifiers. For a finite source with private states X, reports $q : X \to Y$, and lawful labelled transitions $x \xrightarrow{a} x'$, fix an abstract transition relation on Y. Original provenance is a map $\sigma : X \to \Sigma$ preserved by every source edge. Current carrier and epoch may change inside x. Neither this metadata nor a mathematical observation refinement is automatically public.

In the perfect-information realization game, the environment requests an abstract transition $q(x) \xrightarrow{a} y'$. The resolver then chooses a matching private transition $x \xrightarrow{a} x'$ with $q(x') = y'$, before seeing the next request. An unanswered request is a loss. Absence of any abstract request means safe termination, not satisfaction of a progress objective.

5.1 Exact winning regions and finite certificates

It is useful to allow a further adversarial outcome after the resolver's command. Let V be finite; let $\mathrm{Req}(v)$ be its requests, $\mathrm{Resp}(v,r)$ its commands, and $\varnothing \neq \mathrm{Out}(v,r,c) \subseteq V$ their complete outcome sets. Empty response sets express refusal. Define

$$F(Z) = \{v \in V : \ \forall r \in \mathrm{Req}(v)\ \exists c \in \mathrm{Resp}(v,r),\ \mathrm{Out}(v,r,c) \subseteq Z\}.$$

The source-realization game is the special case in which each selected private successor is a response with a singleton outcome set.

Theorem 5.1 (Finite sure-safety certificate)**.** *The greatest fixed point $W = \nu F$ is exactly the sure-winning region, and positional resolver strategies suffice. Its complement has ranks $\rho : V \setminus W \to \{1, \dots, |V|\}$, a chosen request at every losing position, and for every response a chosen lower-rank outcome. These choices force refusal within at most $\rho(v)$ requests from v.*

A positive response table closed in W, together with a ranked negative table on its complement, certifies the exact winning region independently of the algorithm that found it.

Proof. The map F is monotone. Iterating $Z_0 = V$, $Z_{k+1} = F(Z_k)$ gives a decreasing sequence stabilizing after at most $|V|$ strict deletions. Every postfixed set is contained in every Z_k, so its limit is νF. Choose in this limit one response whose complete outcome set remains in the limit for every request. Induction on play length proves sure safety.

Give a state first removed at stage k rank k. Its removal supplies a request for which every response has an outcome outside Z_{k-1}, hence of smaller rank. At rank one there is no response. Following these chosen outcomes strictly decreases a positive integer after every completed round and defeats any history-dependent resolver. The same arguments verify any supplied pair of positive and ranked-negative tables directly. Their regions cannot overlap; if they partition V, they identify the winning region exactly. □

This is classical safety-game and alternating-simulation mathematics; compare Alur, Henzinger, Kupferman and Vardi, *Alternating Refinement Relations* (CONCUR 1998). Its role here is to make control ownership and refusal explicit.

For example, take states p, u, v with one report and initial state p. Either action a, b sends p to either u, v. Only a is available at u, and only b at v; either enabled action again permits both successors. Every finite word has a private lift, since its support becomes $\{u, v\}$ after the first letter and stays there. Yet online ranks are $\rho(p) = 2$, $\rho(u) = \rho(v) = 1$, so there is no online resolver. This is a finite history-determinism obstruction, not a new separation from automata theory; compare Boker and Lehtinen, *Token Games and History-Deterministic Quantitative Automata*, arXiv:2110.14308.

5.2 Public observation requires a command interface

When x is hidden, "choose a successor" is not a public action specification. Declare instead commands $c \in C$ and complete implementation rows $T_{a,c}(x) \subseteq X$, each contained in the lawful a-successors of x. Nature chooses their outcomes adversarially. The public observer sees $h : X \to O$, where h retains q, and has perfect recall of signals, requests, and its own commands. It does not know the original label unless that label is available through h. Beliefs may contain multiple original labels.

At a nonempty homogeneous belief B, the command c answers request (a, y') only if

$$\forall x \in B : \quad \varnothing \neq T_{a,c}(x) \subseteq q^{-1}(y').$$

All actual outcomes must be checked before conditioning: discarding outputs with the wrong report would introduce postselection. After a legal command and observed signal o, update by

$$B_o = \left(\bigcup_{x \in B} T_{a,c}(x) \right) \cap h^{-1}(o), \qquad B_o \neq \varnothing.$$

The belief game has these sets as positions and all nonempty B_o as adversarial command outcomes. It has at most $\sum_o (2^{|h^{-1}(o)|} - 1)$ positions.

Theorem 5.2 (Public sure-safety reduction)**.** *A pure public-history strategy is surely safe from B if and only if B is winning in the belief game. A strategy depending on the current belief and request suffices, although the current report alone may be insufficient.*

Proof. Induction over a public history shows that the belief is exactly the endpoints of all private histories compatible with it. A command must work at each such endpoint, and its possible next signals yield exactly the displayed update. Thus a winning belief strategy yields a lawful public strategy. Conversely, fix a public strategy at a losing belief. The ranked certificate supplies a finite sequence of requests and successor signals leading to a command which is unavailable or has a wrong-report outcome at some compatible endpoint. The whole finite signal history has one compatible private history by the same support induction. Choosing that witness gives a concrete failure of sure safety. This reasoning never switches the realized original source between successive steps. □

This is the safety specialization of the knowledge construction of Chatterjee, Doyen, Henzinger and Raskin, *Algorithms for Omega-Regular Games with Imperfect Information*, LMCS 3(3:4), 2007, Section 3.1. It makes no claim about almost-sure objectives or stochastic belief sufficiency.

Proposition 5.3 (The source relation does not identify public control)**.** *The same private transition relation, initial uncertainty and public reports can have opposite public online winners under different command interfaces.*

Proof. Take indistinguishable initial states $0, 1$, a terminal state d, and edges $0 \xrightarrow{a} d$, $1 \xrightarrow{a} d$. One command implementing both rows is surely successful. Two commands, one available only at 0 and one only at 1, have the same union of private edges but neither is surely legal at belief $\{0, 1\}$. □

5.3 Probe repair: a structural no-go and a hypergraph criterion

Proposition 5.4 (Information cannot repair structural losing states)**.** *If x is losing in the perfect-information private-successor game, no truthful nondestructive probes and public memory can make it surely winning while keeping the source edges and requested abstract relation fixed.*

Proof. A perfect-information private resolver can emulate any such public controller and choose a lawful outcome of each selected command. Sure success of the public controller would contradict the private losing rank certificate. Equivalently, observing more information does not create an edge that escapes the certificate's rank decrease. □

The converse fails by Proposition 5.3. An information defect and a structural control defect require different repairs. A state-changing probe is outside this no-go's assumptions.

For an exact positive result, consider a one-step terminal control problem with one current report and one request. Let $E_x \subseteq C$ be the commands that successfully complete the task at private state x. Let $\mathcal{H}$ contain precisely the inclusion-minimal nonempty sets $H \subseteq X$ with $\bigcap_{x \in H} E_x = \varnothing$. Supply a finite menu of total truthful nondestructive probes $p : X \to O_p$, each with an exact nonnegative cost c_p. A fixed bundle is acquired before the request, and its outputs are retained.

Theorem 5.5 (Minimal fixed-bundle command repair)**.** *A probe bundle S permits a surely successful command if and only if every $H \in \mathcal{H}$ is split by some $p \in S$. Hence minimum acquisition cost is exactly the weighted hitting-set optimum whose probe p covers the hyperedges on which it is nonconstant. A hyperedge split by no available probe certifies impossibility. A singleton hyperedge cannot be repaired by any nondestructive observation.*

Proof. Apply Lemma 4.1 to the hereditary family consisting of the empty set and all nonempty state sets having a common legal command. A hyperedge is monochromatic under the joint probe output exactly when all selected probes are constant on it. Therefore adequacy is precisely the asserted hitting condition, with the same declared additive cost. □

Proposition 5.6 (No coarsest online-safe repair)**.** *Unlike exact autonomous behavior preservation, command-safe observation can have incomparable minimal adequate refinements and no coarsest one.*

Proof. Take $E_0 = \{a, b\}$, $E_1 = \{b, c\}$, $E_2 = \{a, c\}$. Every pair shares a command, but the triple does not. Thus $\mathcal{H} = \{\{0, 1, 2\}\}$: pairwise conflict graphs miss the obstruction. Both partitions $\{0, 1\} \mid \{2\}$ and $\{0\} \mid \{1, 2\}$ are adequate. Any common coarsening identifies $0 \sim 1 \sim 2$, so is constant and inadequate. No coarsest adequate refinement exists. □

This hypergraph mechanism is classical. It prevents equating the coarsest autonomous quotient with a least online-capability repair. The theorem is about supplied fixed probes, not adaptive destructive acquisition.

Verification and authority. The accompanying exact Python solver emits positive strategy and negative rank tables checked by a separate verifier. All 256 two-state/two-command arenas and 60 further three-state/two-request arenas agree with a distinct positional-strategy enumeration oracle. A separate public-policy tree oracle over complete private histories checks 256 public command tables and 768 initial beliefs. These tests validate implementations; the proofs establish the quantified statements. Generic certificate soundness has a separately kernel-checked Lean development in `ForgeGameCertificates.lean`; the fixed-point algorithm and public-belief compiler are not formalized there. No academic priority or physical source authentication is asserted.

6 Authority, prior art, and remaining questions

The stable quotient and coding arguments are classical. Adaptive identification and its failures belong to established finite-state testing [3, 4]; in particular, incomplete distinguishing-sequence covering is close prior art to the feasible-block formulation. Classical original/current state bookkeeping is explicit in that work's Lemma 6.2 and Section 7.1; it is not a new source-indexing invention. Minimum-cost adaptive identification is also established research [6]. Classical zero-error coding connects information with coloring [5]. The present forbidden hyperedges may have size three with no forbidden pairs, which is why a pair-confusability graph alone is insufficient here. Academic novelty of the exact combined cost/source-memory formulation remains unverified. This manuscript should not be described as a new general automata or information theory.

The implementation uses exact rational arithmetic and checks local cost inequalities plus termination ranks. Tests include 160 fresh four-state repair models compared with exhaustive partitions and 90 fresh two-original probe models compared with independent enumeration of stationary policies simulating each fixed original. A complete 13-belief certificate accompanies the three-source

example. A separate raw-table oracle rebuilds beliefs and checks all 6,561 two-carrier probe tables plus 140 fresh three-origin models, including 482 advice-budget comparisons. These are finite implementation checks, not replacement proofs or external source authentication.

The highest-value extensions are a total finite-controller memory bound and a source-preserving restoration requirement after acquisition. Both modify the feasible family and require new arguments. Unbounded carriers, infinite completion, richer proof authority, and physical access remain outside the present theorems.

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